DAEN/ISEN 427 Homework 4 [Practice questions for exam #2]

Due date: Tue Oct 28, 2025 (before class)

Rubric: These questions are provided solely for reference and practice in preparation for the 2nd exam. They will not be graded. Any submission that demonstrates any reasonable Bayesian analysis and a minimum level of effort will receive one point for participation.

1. Compare priors Beta$(0.5, 0.5)$, Beta$(1, 1)$, and Beta$(1, 4)$. How do the priors differ in terms of shape?

   Answer:

2. In the following definition of a probabilistic model, identify the prior, the likelihood, and the posterior:

$$Y \sim \mathcal{N}(\mu, \sigma) \qquad \mu \sim \mathcal{N}(0, 1) \qquad \sigma \sim \mathcal{HN}(1)$$

   $\mathcal{HN}$: the half-normal distribution is a fold at the mean of an ordinary normal distribution with mean zero.

   Hint: $f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

   Answer:

3. In the previous model, how many parameters will the posterior have? Compare your answer with that from the model in the coin-flipping problem in equation:

$$\theta \sim \text{Beta}(\alpha, \beta) \qquad Y \sim \text{Bin}(n = 1, p = \theta)$$

   Answer:

4. Suppose that we have two coins; when we toss the first coin, half of the time it lands tails and half of the time on heads. The other coin is a loaded coin that always lands on heads. If we choose one of the coins at random and observe a head, what is the probability that this coin is the loaded one?

   Answer:

5. Match the following functions that play a role in Bayesian modeling with the descriptions.

   Functions:

   (a) Distribution of the posterior mean estimate
   (b) Prior distribution
   (c) Likelihood function
   (d) Posterior distribution
   (e) Measurement distribution

   Descriptions:

   (a) Is the result of inference on an individual trial
   (b) Describes how potentially noisy observations are generated
   (c) Can be directly compared to human responses in a psychophysical experiment
   (d) Is often modeled as a Gaussian function centered at the measurement
   (e) May reflect statistics in the natural world

   Answer:

6. You need to estimate the weights of blue whales, humans, and mice. You assume the weights are normally distributed, and you set the same prior $\mathcal{HN}(200\text{ kg})$ for the variance (positive scatter parameter). What type of prior is this for adult blue whales? Strongly informative, weakly informative, or non-informative? What about for mice and for humans? How does informativeness of the prior correspond to our real world intuitions about these animals?

   Answer:

7. A student says, "to find the probability distribution of an observer's estimate $\hat{s}$ for a given stimulus s, I can just multiply the measurement distribution $p(x \mid s = \mu)$, where μ is the prior mean, by the prior $p(s)$."

   In other words, the student writes:

$$\text{(wrong)} \quad p(\hat{s} \mid s) = p(x \mid s = \mu)\, p(s).$$

   (a) This formula gives a result that might look reasonable but it is conceptually wrong. Explain why.
   Answer:

   (b) Write down the *correct expression* for the response distribution $p(\hat{s} \mid s)$ in terms of the measurement distribution $p(x \mid s)$ and the observer's decision rule $p(\hat{s} \mid x)$. Briefly explain what each term represents.
   Answer:

   (c) Sketch or describe how the *correct* response distribution $p(\hat{s} \mid s)$ would change if the measurement noise increased. Hint: think about how $p(x \mid s)$ becomes wider and how that affects the spread of responses.
   Answer:

8. In the *Bayesian perception framework*, an observer tries to estimate the stimulus s based on a noisy measurement m.

   - $p(m \mid s)$: *Measurement distribution* — probability of observing measurement m given stimulus s.
   - $p(s)$: *Prior* — how likely different stimuli are before seeing the data.
   - $p(s \mid m)$: *Posterior* — probability of each stimulus given the measurement.
   - $p(m)$: *Evidence (marginal likelihood)* — overall probability of the measurement.
   - $p(r \mid s)$: *Response distribution* — probability that the observer gives response r for stimulus s.

   These quantities are related through Bayes' rule:

   $$p(s \mid m) = \frac{p(m \mid s)\, p(s)}{p(m)}, \quad p(r \mid s) = \int p(r \mid m)\, p(m \mid s)\, dm.$$

   The goal is to understand how the following concepts are related: (*i*) likelihood function, (*ii*) measurement distribution, (*iii*) posterior, (*iv*) prior, and (*v*) the observer's response or estimate.

   True or False? If false, please explain why.

   (a) The likelihood function is always equal to the measurement distribution.
   Answer:

   (b) The stimulus value that maximizes the posterior probability is the same as the measurement value.
   Answer:

   (c) The response distribution for a given stimulus can be found by multiplying the measurement distribution by the prior.
   Answer:

   (d) The response distribution for a given stimulus is always Gaussian.
   Answer:

   (e) If a Bayesian observer reports one stimulus value more often than another, it means the prior probability of that value is higher.
   Answer:

9. In Bayesian inference, we often want to compute the posterior distribution

$$p(\theta \mid D) = \frac{p(D \mid \theta)\, p(\theta)}{p(D)},$$

where θ are model parameters and D is observed data. In most realistic problems, computing the denominator

$$p(D) = \int p(D \mid \theta)\, p(\theta)\, d\theta$$

is computationally heavy (or intractable because it involves integrating over a high-dimensional parameter space).

(a) Explain why the normalization constant $p(D)$ is difficult to compute.
Answer:

(b) Describe one computational method that can be used to approximate the posterior $p(\theta \mid D)$ without evaluating $p(D)$ directly. Give an example of how this method works in principle.
Answer:

(c) Suppose we use a *Metropolis–Hastings* Markov Chain Monte Carlo (MCMC) algorithm to sample from $p(\theta \mid D)$. Briefly describe how a proposed sample θ' is accepted or rejected.
Answer:

Answer:

(d) Discuss one advantage and one disadvantage of using MCMC.
Answer:

10. Suppose we are estimating a parameter θ that represents the mean of a Gaussian process. We observe one noisy measurement $x = 5.2$ drawn from

$$p(x \mid \theta) = \mathcal{N}(x; \theta, \sigma^2), \quad \text{where } \sigma^2 = 1.$$

We assume a Gaussian prior on θ:

$$p(\theta) = \mathcal{N}(\theta; 0, \tau^2), \quad \text{where } \tau^2 = 4.$$

Hint:

- For a Gaussian likelihood and prior, the posterior is also Gaussian with:

$$\mu_{\text{post}} = \frac{\frac{x}{\sigma^2} + \frac{0}{\tau^2}}{\frac{1}{\sigma^2} + \frac{1}{\tau^2}}, \quad \sigma^2_{\text{post}} = \frac{1}{\frac{1}{\sigma^2} + \frac{1}{\tau^2}}$$

- The unnormalized posterior is proportional to $\exp\left[-\frac{1}{2\sigma^2}(x-\theta)^2 - \frac{1}{2\tau^2}\theta^2\right]$
- $r = e^{\log r}$

(a) Write down the unnormalized posterior distribution $p(\theta \mid x) \propto p(x \mid \theta)p(\theta)$.
Answer:

(b) Calculate the numerical values of μ_{post} and σ^2_{post} for the given parameters.
Answer:

(c) Now suppose we do not know the analytical form of the posterior and want to use the *Metropolis–Hastings algorithm* to sample from $p(\theta \mid x)$. Starting at $\theta_t = 2.0$, we propose a new value $\theta' = 3.0$ from a symmetric proposal distribution. Compute the acceptance probability

$$\alpha = \min\left(1, \frac{p(x \mid \theta')\, p(\theta')}{p(x \mid \theta_t)\, p(\theta_t)}\right).$$

Show all steps and give the numerical value of α.
Answer: